

Packet Tracer Lab Guide: Fog and Edge Computing

A structured lab that models outer-edge devices, a middle-edge fog node, and a cloud back end to demonstrate why computation moved closer to data sources improves latency, continuity, and efficiency.

Suggested duration	90-120 minutes
Core lecture concepts modelled	BLURS, SCALE, SCANC, and the inner-edge / middle-edge / outer-edge hierarchy

Important modelling note: Cisco Packet Tracer can represent the placement of control, storage, and networking functions very well, but it does not emulate full container orchestration, GPU/FPGA acceleration, or production fog platforms. This lab therefore implements the concept through service placement, automation rules, path length, and resilience testing.

1. Lab Overview

This lab builds a three-layer IoT network in Packet Tracer that approximates the fog and edge computing model described in the lecture. You first create a cloud-connected IoT environment and then place control and processing functions closer to the devices, so that the system can respond locally, keep operating during WAN failure, and reduce unnecessary dependence on the remote cloud.

The lab is intentionally grounded in four ideas from the lecture: the cloud-centric IoT model suffers from BLURS challenges; fog and edge computing reduce latency and improve efficiency; FEC capabilities can be explained through SCANC; and the architecture can be understood as outer-edge, middle-edge, and inner-edge layers. In the simulation, the outer-edge contains sensors and actuators, the middle-edge contains the local gateway and fog server, and the inner-edge contains the remote cloud server.

2. Scenario

Scenario used in this guide: a smart warehouse monitors temperature and motion. The warehouse must switch on a local fan when temperature rises above a threshold and must raise an alarm when motion is detected in a restricted area. Recent operational data should remain visible on a nearby fog node, while the cloud receives only a summarized or backup view.

This scenario suits Packet Tracer because it shows the difference between cloud-only control and distributed control. If the WAN path fails, local automation must continue to work. That directly demonstrates why the lecture argues that fog and edge computing are useful for latency-sensitive, bandwidth-aware, and intermittently connected IoT systems.

3. Learning Outcomes

- Explain the difference between cloud computing, edge computing, and fog computing in an IoT topology.
- Map Packet Tracer components to outer-edge, middle-edge, and cloud/inner-edge roles.
- Configure routing, addressing, and server placement so that local processing happens nearer the data source.
- Demonstrate that local control still works when the WAN link to the cloud is unavailable.
- Relate the finished design to BLURS, SCALE, and SCANC from the lecture.

4. Concept-to-Lab Mapping

Lecture concept	Meaning in the lecture	How the lab models it
BLURS	Bandwidth, latency, uninterrupted operation, resource constraints, and security challenges in cloud-centric IoT.	The lab compares cloud-only dependence with local automation and local visibility; WAN failure tests expose the value of uninterrupted local control.
SCALE	Security, cognition, agility, latency, and efficiency benefits of FEC.	Latency and efficiency are shown by keeping responses local; cognition and agility are shown by placing decisions on local nodes; security can be strengthened with an optional ACL.
SCANC	Storage, compute, acceleration, networking, and control are the core mechanisms of FEC nodes.	Fog server stores recent data, gateway and fog node perform control/compute, IP/Wi-Fi provide networking, and automation rules

Lecture concept	Meaning in the lecture	How the lab models it
		plus ACLs implement control. Acceleration is approximated by the shorter decision path rather than special hardware.
Hierarchy of FEC	Outer-edge = devices; middle-edge = local LAN/cellular fog resources; inner-edge = wider network and cloud connectivity.	Sensors and actuators are the outer-edge; the local gateway and fog server are the middle-edge; the remote server is the cloud/inner-edge back end.

5. Devices and Materials

- Packet Tracer is available via Appsanywhere. Device names vary slightly by version; use equivalent IoT or server devices where needed.
- 2 routers (or one router plus one cloud/ISP router), 3 switches, 1 home gateway or wireless access gateway, 1 fog server, 1 cloud server, 1 admin PC.
- At least 2 sensors and 2 actuators. A temperature sensor plus fan, and a motion sensor plus alarm or door lock, work well.
- Copper straight-through cables for LAN links and a serial/WAN link between the site router and the cloud router.

6. Recommended Topology

Build the topology in three zones from left to right so the architecture is easy to read on the workspace: outer-edge devices on the left, the middle-edge fog layer in the center, and the cloud on the right.

Outer-edge zone	Middle-edge / fog zone	Inner-edge / cloud zone
Temperature sensor Motion sensor Fan or alarm actuator Home Gateway / IoT Gateway Optional local edge PC	SW-FOG Fog Server R-EDGE Local web/IoT service Optional cached logs	R-CLOUD Cloud Server Admin PC Remote dashboard Backup / long-term storage

7. Addressing Plan

Network	Purpose	Example subnet	Key devices
Outer-edge LAN	IoT devices and local gateway	192.168.10.0/24	R-EDGE G0/0 = 192.168.10.1, Home Gateway = 192.168.10.2
Fog LAN	Local server and aggregation	192.168.20.0/24	R-EDGE G0/1 = 192.168.20.1, FOG-NODE = 192.168.20.10
Cloud LAN	Remote services and admin access	192.168.30.0/24	R-CLOUD G0/0 = 192.168.30.1, CLOUD-DC = 192.168.30.10

Network	Purpose	Example subnet	Key devices
WAN link	Site-to-cloud transport	10.0.0.0/30	R-EDGE S0/0/0 = 10.0.0.1, R-CLOUD S0/0/0 = 10.0.0.2

8. Detailed Packet Tracer Procedure

The steps below assume a common Packet Tracer workflow. Menu names may differ slightly across versions, so keep the same logic even if a button label or service pane is not identical on your installation.

Step 1 - Place and name the devices

1. Drag the routers, switches, servers, PC, home gateway, sensors, and actuators onto the workspace.
2. Arrange them physically in the three zones shown earlier so the architecture communicates the FEC hierarchy clearly.
3. Rename the main devices as R-EDGE, R-CLOUD, SW-EDGE, SW-FOG, SW-CLOUD, FOG-NODE, CLOUD-DC, and ADMIN-PC.

Step 2 - Cable the topology

4. Connect the home gateway and any local edge PC to SW-EDGE with copper straight-through cables.
5. Connect FOG-NODE to SW-FOG and CLOUD-DC plus ADMIN-PC to SW-CLOUD.
6. Connect SW-EDGE and SW-FOG to R-EDGE using separate Gigabit interfaces.
7. Connect R-EDGE to R-CLOUD using a serial link or another WAN link type available in your Packet Tracer setup.
8. Connect SW-CLOUD to R-CLOUD.

Step 3 - Configure router interfaces and routing

Apply the following sample configuration. Adjust interface numbers if your router model uses different names.

```
R-EDGE
conf t
hostname R-EDGE
interface g0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
interface g0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
interface s0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown
ip route 192.168.30.0 255.255.255.0 10.0.0.2
```

```
R-CLOUD
conf t
hostname R-CLOUD
interface g0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
interface s0/0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
ip route 192.168.10.0 255.255.255.0 10.0.0.1
ip route 192.168.20.0 255.255.255.0 10.0.0.1
```

After configuring the routers, verify basic IP connectivity by pinging between the router interfaces. Do this before touching the IoT services so the underlay network is known to be correct.

Step 4 - Configure the fog and cloud servers

9. Set FOG-NODE to 192.168.20.10/24 with default gateway 192.168.20.1.
10. Set CLOUD-DC to 192.168.30.10/24 with default gateway 192.168.30.1.
11. Set ADMIN-PC to 192.168.30.20/24 with default gateway 192.168.30.1.
12. On both servers, enable HTTP so students can browse to the local fog page and the remote cloud page.
13. If your Packet Tracer version supports IoT registration on a server, enable the IoT service on FOG-NODE and create a local account such as warehouse / warehouse123.

Step 5 - Configure the outer-edge gateway and wireless settings

14. Give the Home Gateway the address 192.168.10.2/24 with default gateway 192.168.10.1.
15. Set a clear SSID such as EDGE-IOT and apply a simple WPA2 key for the lab.
16. If using local registration, point the gateway or IoT registration function toward the fog server at 192.168.20.10.
17. Record the SSID and key in the students' worksheet so they can onboard sensors consistently.

Step 6 - Add and onboard the sensors and actuators

18. Open each IoT device and join it to the EDGE-IOT wireless network.
19. Name the devices clearly, for example TEMP-SENSOR, MOTION-SENSOR, FAN-ACTUATOR, and ALARM-ACTUATOR.
20. Where available, register the devices to the local fog/IoT service account created earlier.
21. Test that the devices show as online before proceeding.

If your version does not expose IoT registration exactly as described, keep the same structure by joining the devices to the local gateway and using the gateway as the local edge controller. The conceptual goal is unchanged: the first decision point must be close to the devices, not in the remote cloud.

Step 7 - Implement local edge logic

This is the most important step conceptually. Configure the gateway or local control pane so that the first response happens at the edge. Use built-in conditions, rules, or programming blocks available in your Packet Tracer version.

22. Create Rule 1: If temperature exceeds a threshold, turn FAN-ACTUATOR on.
23. Create Rule 2: If motion is detected, turn ALARM-ACTUATOR on or lock a smart door.
24. Keep these actions local. Do not make the WAN or cloud server a requirement for the immediate response.

This step models edge computing from the lecture: computation and control are placed at or near the device subnet, so critical responses do not depend on the distant cloud.

Step 8 - Implement the fog function

25. Use FOG-NODE as the nearest server that students and administrators can reach over the LAN.
26. Create a simple local status page on FOG-NODE describing the current warehouse state, such as latest temperature, motion state, and actuator state.
27. Treat FOG-NODE as the local aggregation, storage, and management point. In the report, explicitly describe this as the middle-edge layer.
28. If you want to strengthen the simulation, add a note or log file on FOG-NODE representing cached or recent operational data.

In a real FEC deployment, this node could host containers, caching, local analytics, or protocol translation. Packet Tracer approximates that behaviour through local service placement rather than true virtualization.

Step 9 - Implement the cloud function

29. On CLOUD-DC, enable HTTP and create a page labelled Remote Cloud Dashboard or Backup Service.

30. Use this server to represent long-term storage, remote visibility, or less time-critical processing.

31. From ADMIN-PC, verify that the cloud page is reachable over the WAN.

At this point the finished architecture should express a clear division of roles: immediate control at the edge, local aggregation at the fog layer, and backup or enterprise visibility in the cloud.

Step 10 - Optional security and mediation control

To demonstrate the control function from SCANC, you can prevent outer-edge devices from talking directly to the cloud and force them to use the fog layer as the local mediator.

```
R-EDGE
conf t
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 permit ip any any
interface g0/0
 ip access-group 110 in
```

Apply this only after the basic lab works. The ACL is pedagogically useful because it illustrates that fog nodes can mediate communication and reduce unnecessary direct dependence on the cloud.

Step 11 - Verification tasks and experiments

Test	Action	Expected result	Lecture link
1	Ping from outer-edge LAN to fog server and cloud server.	Both respond while WAN is up.	Basic vertical connectivity
2	Change the temperature value so the threshold is exceeded.	Fan turns on locally without needing a cloud action.	Low latency / edge control
3	Trigger motion detection.	Alarm or door action occurs locally.	Real-time edge response
4	Shut down the WAN interface between R-EDGE and R-CLOUD.	Cloud becomes unreachable, but local automation still works and fog resources remain reachable.	Uninterrupted service
5	Re-enable the WAN link and browse from ADMIN-PC to both fog and cloud pages.	Normal operation resumes and remote view is restored.	Hybrid FEC + cloud
6	If ACL is used, test direct edge-to-cloud traffic.	Outer-edge traffic is blocked from going directly to cloud.	Control / security / mediation

9. Expected Interpretation of Results

- When the fan or alarm reacts locally, the lab demonstrates why edge computing is suited to time-sensitive functions.
- When the fog server remains reachable during WAN failure, the lab demonstrates the lecture argument that FEC reduces dependence on uninterrupted long-distance connectivity.
- When the cloud is used mainly for remote oversight or backup, the design reflects the lecture distinction between local fog resources and distant cloud resources.
- When an ACL forces local mediation, the lab demonstrates the control and security dimension of SCANC.

10. Troubleshooting Guide

Problem	Likely cause	Fix
No device can reach the fog server	Default gateway or router interface error	Check the IP addressing table, verify interface status, and use ping hop-by-hop.
IoT device is offline	Wrong SSID, password, or registration target	Reconnect the device to EDGE-IOT and verify the fog server address or local gateway settings.

Problem	Likely cause	Fix
Local rule does not fire	Condition not saved or wrong threshold	Re-open the automation/conditions pane and test with an obviously high or low sensor value.
Cloud test fails only after ACL	ACL placed on wrong interface or too restrictive	Remove the ACL, confirm the baseline works, then reapply carefully.
Fog concept is not obvious in the final topology	All services are treated as equally remote	Visually separate the three zones and explain each role explicitly in labels and notes.

11. What You Should Submit

- The final Packet Tracer .pkt file.
- A screenshot of the completed topology with labels for outer-edge, middle-edge/fog, and cloud.
- One screenshot proving local actuation works.
- One screenshot proving fog access remains available when the WAN link is down.
- Short written answers to the reflection questions below.

12. Reflection Questions

1. Which BLURS challenge is most clearly reduced by the local edge rule in this lab?
2. What part of your topology is acting as the middle-edge fog resource, and why is it not the same thing as the remote cloud?
3. Which elements of SCANC are visible in your implementation, and which ones can only be approximated in Packet Tracer?
4. How would this design need to change for a smart hospital or autonomous vehicle scenario where latency is even more critical?
5. Why is it useful to keep the cloud in the design even after adding fog and edge functions?

13. Possible Extensions

- Scale the lab by adding another home gateway and a second fog-facing subnet to represent multiple local zones feeding one fog server.
- Replace the warehouse context with smart healthcare, smart farming, or connected vehicles while keeping the same architectural pattern.
- Use Simulation Mode to show that local control traverses fewer hops than a cloud-dependent workflow.

14. Conclusion

This Packet Tracer exercise does not try to imitate every engineering detail of a real fog platform. Instead, it makes the architecture visible: sensing happens at the edge, immediate decisions happen locally, nearby fog resources provide local aggregation and resilience, and the cloud remains available for remote access and longer-term processing.